Omar AbedelKader

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Interests: Artificial Intelligence · Software Engineering · LLMs · AI4SE

About Me

Biography:

- I am an AI researcher with extensive experience in artificial intelligence and software engineering.
- I am currently pursuing a Ph.D. in Computer Science at the University of Lille.
- I am currently involved in several open-source, research, AI, and community initiatives, where I contribute as a founder, maintainer, project leader, president, and active member.
- You can see the full list of my current responsibilities **here**.

Awards:

- Best Paper Award (3rd place) — IWST 2025, Gdańsk, Poland
- Alumni Testimonial — IDMC 2024, University of Lorraine, France *

Publications

Rank A

- Kier, K., Giagnorio, A., Abedelkader, O., Zaitsev, O., Peharz, R., Robbes, R., Bavota, G., & Ducasse, S. (2026). Teaching LLMs a Low-Resource Language: Enhancing Code Completion in Pharo. *International Conference on Software Maintenance and Evolution (ICSME)*. DOI.

Rank C

- AbedelKader, O., Ducasse, S., Zaitsev, O., Robbes, R., & Polito, G. (2025). Package-Aware Approach for Repository-Level Code Completion in Pharo. *International Workshop on Smalltalk Technologies (IWST)*. DOI.
- Elkolei, M., AbedelKader, O., Ducasse, S., & Zaitsev (2026). Alternative UX Extensions and Their Trade-offs for Code Completion in Pharo. *International Workshop on Smalltalk Technologies (IWST)*. DOI.

Q2 Journals

- AbedelKader, O., Ducasse, S., Zaitsev, O., Robbes, R., & Polito, G. (2026). Evaluating Package-Level Scoping Strategies for Repository-Level Code Completion in Pharo. *Journal of Computer Languages (COLA)*. DOI.

Services

2027

- Committee Member in Artifact Track at FormaliSE'27
- Shadow Reviewer at ICSE'27
- Contributor at Café IA (Conseil National du Numérique).
- Member at Club IA et Transition écologique (Ministère de la Transition Écologique).
- Website Maintainer GDR-SciLog
- Project Leader and Maintainer for multiples projects (ChatPharo, Pharo-Infer and many more).
- Member in the organization Pharo-AI.
- Founder and Maintainer of the organization Pharo-LLM.

2026

- Member of the organizing committee, GDR-SciLog'26.
- Member of the organizing committee, ESUG'26.
- Contributor at Café IA (Conseil National du Numérique).
- Member at Club IA et Transition écologique (Ministère de la Transition Écologique).
- President of Synapse-NeuroTech-Lille, a student association focused on AI and neuroscience.
- Website Maintainer GDR-SciLog
- Project Leader and Maintainer for multiples projects (ChatPharo, Pharo-Infer and many more).
- Member in the organization Pharo-AI.
- Founder and Maintainer of the organization Pharo-LLM.

2025

- Member of the organizing committee ESUG'25
- Sub reviewer at BENEVOL'25.
- Contributor at Café IA (Conseil National du Numérique).
- Member at Club IA et Transition écologique (Ministère de la Transition Écologique).
- Project Leader and Maintainer for multiples projects (ChatPharo, Pharo-Infer and many more).
- Member in the organization Pharo-AI.

- Founder and Maintainer of the organization Pharo-LLM.

2024

- Member of the organizing committee ESUG'24.
- Contributor at Café IA (Conseil National du Numérique).
- Member at Club IA et Transition écologique (Ministère de la Transition Écologique).

2023

- Volunteer First Aider in the French Red Cross

Teaching

2027

- Teaching assistant at IUT : *Software Maintenance* 12h Tutorials IUT-Lille
- Teaching assistant at Polytech: *DataBase* 16h Labs Polytech-Lille
- Teaching assistant at Polytech: *Internet* 12h Labs Polytech-Lille

2026

- Teaching assistant at IUT: *Introduction to Software Development* 41,5 h Labs IUT-Lille
- Teaching assistant at IUT : *Software Maintenance* 12h Tutorials IUT-Lille
- Teaching assistant at Polytech: *DataBase* 16h Labs Polytech-Lille
- Teaching assistant at Polytech: *Internet* 12h Labs Polytech-Lille

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Software

ChatPharo

- Live conversational interface between Pharo developers and large language models.
- Enables interactive discussions with LLMs directly from the Pharo environment.
- Designed an open and extensible architecture to experiment with multiple LLM backends.
- GitHub Repository: ChatPharo

INERIS-IA

- AI platform developed at INERIS for document classification and knowledge management.
- Built a Flask-based web interface integrating multiple ML and NLP models.
- Trained document classification models aligned with COP 2027 objectives and INERIS strategic themes.
- Implemented additional features including document similarity, keyword extraction, and corpus quality improvement.
- GitHub Repository: INERIS-IA

Students

2026

- Mehdi Elkolei, Bachelor Intership, 4 month, April 2026 - July 2026
  I supervised Mehdi Elkolei on his work on extending Complishon, Pharo's context-aware code completion engine. His project focused on four UX-oriented extensions: typo tolerance, implicit prefix expansion, grouped completion entries, and camel-case matching. I guided the technical direction and design choices, with particular attention to usability, modularity, and integration with the existing architecture. Elkolei, M., AbedelKader, O., Ducasse, S., & Zaitsev (2026). Alternative UX Extensions and Their Trade-offs for Code Completion in Pharo. *International Workshop on Smalltalk Technologies (IWST)*. DOI.

2025

- Srajal Dwivedi, Google Summer of Code Intership, 3 month, May 2025 - July 2025 I supervised Srajal the ChatPharo project, which aimed to integrate LLMs into the Pharo IDE. It was a Google Summer of Code 2025 project. Srajal improved ChatPharo's user interface and implemented new features to facilitate interaction with LLMs. The project was well-received by the Pharo community and contributed to the adoption of ChatPharo by developers. It was also one of the first projects to integrate LLMs into the Pharo IDE.

Public Talks

2026:

- ESUG 2026: ChatPharo: an AI Assistant inside Pharo
- ESUG 2026: Pharo-LLM: Building a sovereign LLM ecosystem in Pharo
- French Brain Awareness Week: AI and the brain: two ways of learning, what does that change for our daily lives?

2025:

- ESUG 2025: ChatPharo: An Open Architecture for Understanding How to Talk Live to LLMs, Gdansk, Poland
- GT GL: ChatPharo: Talking Live to LLMs from Pharo, Lille, France
- Scientific Seminar: LLM101-2, Lille, France

- Scientific Seminar: LLM101, Lille, France

Professional Experience

- PhD Student - Oct. 2024 - Oct. 2027 - INRIA Lille / LaBRI Bordeaux
- Invited Researcher - Sep.2026 - Dec.2026 - Polytechnique Montréal - Montréal, Canada
- AI Engineer - Sept. 2023 - Sept. 2024 - Institut national de l'environnement industriel et des risques (INERIS) - Verneuil-en-Halatte, France
- Intern - June. 2023 - Aug. 2023 - Laboratoire d'Informatique de Paris Nord (LIPN) - Villetaneuse, France

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Education

- Ph.D.in Computer Science, University of Lille
- M.Sc. in Natural Language Processing, University of Lorraine
- B.Sc. in Data Science, Lebanese University

Hobbies

- **Water Activities:** Swimming · Diving
- **Outdoor and Adventure Sports:** Hiking · Skiing · Skydiving
- **Creative and Artistic Activities:** Photography
- **Precision Sports:** Boxing · Bowling · Golf

Languages

- **French:** Native
- **English:** B2
